

MIRAZ HOSSAIN

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Education

- **Ahsanullah University of Science and Technology (AUST)** Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering(EEE) ; CGPA: 3.024/4.00 Nov 2021 - Nov 2025
Relevant Courses:Circuit Design, Power System, Control system design, MCU, Computer Architecture, Digital Signal Processing, Mechanics of Material, Optical Communication, Fluid mechanics , C++ Programming, VLSI, Image Processing, Medical Device design, C A D
- **Dhaka Residential Model College** 2020
Higher Secondary School Certificate (HSC); Science ; GPA: 5.00/5.00
- **Pabna Zilla School** 2018
Secondary School Certificate (SSC); Science ; GPA:5.00/5.00

Research Interests

- Robotics & Automation • IoT • Embedded Systems design • Machine Learning • Communication Protocols • Circuit and PC B Design • Android App Development • VLSI Design • Deep Learning

Thesis

- AI Adaptive PID Controller – Developed a system to control the motion of a robotic arm with adaptive AI algorithms

Management experience

- **AUST Innovation and Design Club (IDC)**
Executive Member & Assistant Head of Project Team
 - Led technical projects, fostering innovation and effective team collaboration.
- **AUST Satellite Lab**
Founding Member
 - Supported satellite research with a focus on embedded systems design and testing.
- **AUST Model Rocketry**
Software and Navigation Team Member
 - Contributed to model rocketry system design, testing, and automation.

Work Experience

- **Cybernetics Hi-Tech Solution (Pvt.) Ltd.**
Embedded System Engineer – R N D Intern (Feb 2026 - present)
 - Responsible for verifying the technical decisions made by team members, ensuring accurate implementation and compliance with industry standards.
 - Contributing to the design and development of **Autonomous Guided Vehicles (AGVs)** for agricultural and factory applications, optimizing both hardware and software performance.
 - Involved in the study of Meter Communication protocols such as **DCU, AMI, and billing methods**. Focused on enhancing data transmission efficiency and reliability in metering systems.
 - Conducting in-depth studies on **STS** cryptography and its implementation in the **DLMS** protocol for secure metering solutions.
 - Experimenting with the implementation of a 20-digit token system for meters to enhance security and reliability in utility metering.
 - Currently programming **Cortex M0**-based microcontrollers to design and implement anti-theft protection systems, leveraging low-power consumption and high-performance features.
 - Involved in calibrating **PLC** modules for seamless integration and data transmission over power lines, ensuring stable and accurate performance.
 - Contributing on integrating **ROS (Robot Operating System)** and **Micro-ROS** with embedded systems, focusing on communication protocols and real-time processing in robotics.
 - Contributing object tracking systems for drones, enabling communication between cameras, sensors and **Raspberry Pi** for real-time object identification and navigation.

Logistics experience

- **Supply Chain:** Sourcing of raw materials from local markets e.g. Jonson Road, Dholaikhal, Patuatoli Nawabpur, Wari, Sadarghat and many obscure locations.
- **Supply Chain:** Global sourcing experience: International - Digikey, JLC PCB etc . Chinese -Alibaba, Made-in-China, AliExpress etc.
- **Negotiation:** Experience in price negotiation on chinese sourcing websites.
- **Finance:** Experience in R&D and product development budget compilation.

Undergrad Projects

- **Computer Vision Gesture-Controlled Lighting:** Designed a custom PCB for a hand-gesture recognition system using **OpenCV** and **MediaPipe** to facilitate touchless HMI (Human-Machine Interface) for load control.
- **16-DOF Humanoid Robot & Android Interface:** A 16-Degree-of-Freedom robot using **AutoCAD** for 2D frame design and laser-cut fabrication; developed a custom **Android application** for real-time kinematic control and movement synchronization.
- **Advanced Overcurrent Protection & GSM Diagnostics:** Developed an intelligent power protection system featuring **GSM-based fault reporting** and automated diagnostic alerts for electrical grid safety.
- **Vocal Navigation & Speech-Triggered Vehicle:** Architected a voice-actuated control system for a mobile robot, integrating **Natural Language Processing (NLP)** to translate speech commands into hardware execution sequences.
- **IoT Health Monitoring & Personal Safety System:** A dual-purpose portable system for hazardous environments, integrating **GPS tracking**, environmental sensors, and real-time health telemetry via IoT protocols.
- **Self-Balancing Breadboard Platform:** Implemented a real-time stabilization system utilizing a **Kalman Filter** for sensor fusion and a **PID Control** algorithm to maintain equilibrium on a breadboard-based chassis.
- **Distributed Smart Home Framework :** Developed multi-platform automation ecosystem using **Node-RED**, **ESP-RainMaker**, and **SinricPro** for seamless, low-latency integration of household appliances and smart loads.
- **Precision Agricultural Climate Optimizer:** Integrated a multi-sensor IoT node to monitor and adjust micro-climates in farming environments, focusing on **environmental optimization** through data-driven feedback loops.
- **Spectral Signal Analysis (Burg, Yule-Walker, Levinson-Durbin):** Performed advanced frequency estimation and digital signal processing on speech signals using parametric modeling techniques including **Burg's Method** and **Levinson-Durbin recursion**.
- **RFID-Based Intelligent Access & Attendance System:** Designed a secure, real-time tracking system using **RFID technology** for automated attendance logging and encrypted entry management.
- **Autonomous Navigation Suite (Gyro, Line, Human, & Obstacle):** Developed a series of mobile robotic platforms including a **Gyro-controlled vehicle**, **Human-following** bot, **Line-follower**, and **Obstacle-avoidance** system using ultrasonic and IR sensor arrays.
- **Industrial Smart Color Classification System:** An automated sorting system for manufacturing lines using high-speed color sensors and logic-based actuation for quality control.
- **NTP-Server & RTC Based Smart Digital Clock:** Fabricated a network-synchronized clock using an **ESP8266**, **MAX7219** display, and a custom **3D-modeled and printed** chassis, ensuring millisecond accuracy via NTP protocols.
- **IoT Smart Waste Management (ThingSpeak):** Developed a "Smart Dustbin" featuring **ultrasonic fill-level monitoring** and air quality/dust sensors with real-time data visualization on the **ThingSpeak cloud**.
- **CanSat Weather Monitoring System:** Designed and deployed a telemetry-based "CanSat" for atmospheric data acquisition, focusing on real-time weather parameter transmission and analysis.
- **Digital AC Voltage & Frequency Instrumentation:** A high-precision digital meter to monitor mains power quality, measuring frequency and voltage fluctuations in real-time.
- **Robotic Arm IR-Control & Automation:** Developed a multi-axis robotic arm with an **IR-remote interface** for manual maneuvering and programmed sequence execution.
- **Interactive Welcoming Robot with ID Verification:** A "talking" robot capable of vocal interaction and security-based entry validation via ID card scanning.
- **Python-Based Geospatial Geocoding App:** Created a Python application for automated geocoding to extract location data and mapping coordinates from phone numbers.
- **ESP32-Based Air Mouse (HID):** Developed a wearable Human Interface Device (HID) using an **ESP32** and **IMU sensor** to control PC cursor movement through spatial gestures.
- **Machine Learning Diabetes Predictive Model:** Developed and trained a neural network using **TensorFlow** to perform clinical data analysis and binary classification for diabetes risk assessment.
- **45nm CMOS Full Adder Design (VLSI):** Conducted high-level circuit analysis using **Cadence Virtuoso** to calculate and optimize Delay, Power, Power-Delay Product (PDP), and Energy-Delay Product (EDP) for a 45nm technology-based Full Adder.

- **Automated Fluid Management System:** Implemented a sensor-driven water management system with automated alarm triggers and pump control to prevent overflow and optimize usage.
- **Security-Enhanced Smart Door Lock:** Designed a biometric or multi-feature digital lock system focusing on advanced encryption and hardware-level security protocols.

Skills Summary

- **Single Board Computer :** Raspberry Pi 4b
- **MicroControllers:** Arduino Uno, Arduino Nano, Arduino Mega, ESP32, ESP8266, ESP32C3, STM32, Raspberry Pi Pico, HT5023
- **Electrical Design:** Altium Designer, EasyEDA, Cadence Virtuoso , Proteus , Quartus Prime , Simulink , Orcad Pspice
- **Protocols:** UART, SPI, I²C, USB (CDC), Bluetooth, Wi-Fi, Ethernet, GSM/GPRS (AT Commands), GPS (NMEA), RS485, LoRa, MQTT, HTTP/HTTPS, TCP/IP
- **Software and tools:** C+ +, Python, Verilog, Git, Matlab and Simulink, Arduino IDE, Thonny IDE, STM32 Cube IDE, Keil IDE, Visual Studio Code, PlatformIO, Micropython, OpenCV, Mediapipe, Numpy, Tensorflow, YOLO model, DeepSORT, Android Studio, MIT App Inventor Computer Networking etc.
- **Hardware and tools:** **Environmental & motion sensors** (DHT11, BMP180, BMP280, MPU6050, ADXL335, ADXL 345, GY BNO055, GP2Y0A21YK0F, GP2Y1010AU0F, HC-SR04 , KY-018 LDR, KY-026 Flame Sensor, TCRT5000 IR Sensor, TTP223 Touch Sensor, SW-420 Vibration Sensor, TCS230 TCS3200 Color Sensor, MQ-4 Gas Sensors), **positioning & navigation** (NEO-6M GPS, NEO-7M GPS, EC11 Rotary Encoder), **wireless & communication modules** (SX1278 LoRa, NRF24L01+, HC-05, HC-06, SIM808, SIM800L, SIM900A GSM/GPRS, SIM A7670C GSM Module, TSOP1738 IR Receiver, ASK Module 433Mhz, MNL80A PLC module, FT232RL FTDI USB-UART, ST link V2, USBasp,), **motor control& actuation** (28BYJ-48 Stepper, NEMA-17 Stepper, SG90 Servo, MG996R Servo, BLDC Motors, ESC 30A, Industrial Servo, TB6600 Stepper Motor Driver, L293D, L298N, TB6612FNG, BTS7960, Cytron MD10C, PCA9685), **power & energy systems** (DALY BMS, LM2596 Buck, mini360 Buck converter, MT3608 Boost, LDO Ics, ZMPT101B Voltage Sensor, ACS712 Current Sensor, PZEM-004T, PCF8574 IO Expander Module B6 Li-Po Charger,), **measurement & industrial sensors** (YF-S201 Flow Sensor, HX711 Load sensor, MAX30100 Pulse Sensor, FC-28 Water Level Sensor, FC-28 Soil Moisture Sensor, HC-SR501 PIR Motion Sensor), **displays & HMI** (TM1637 7-Segment, 16×2 LCD, SSD1306 OLED, MAX7219 Display Driver, ST7789V driven SPI TFT LCD Display, 4×4 Matrix Keypad, KY-023 Joystick), **storage & identification** (DS3231 RTC, MicroSD Card Module, MFRC522 RFID), **vision control & advanced systems** (ESP32-CAM OV2640, Pixhawk PX4 2.4.8 Flight Controller, FlySky FS-i6X Remote, PAM8610 Audio Amplifier).
- **CAD Software:** SolidWorks , AutoCad.
- **Documentation tools :** Microsoft Office Suite (Word, Excel, PowerPoint)

Honors and Awards

- Received Invitation Letter to participate in Cozmoclench, TechFest IIT Bombay – 2022
- Srijonshil Medha Onneshon (2nd position in Rajshahi Devision, High School, Class-09, 2016)

Extracurricular activities

- Played soccer in club.
- Playing Football.

References

- Mr. Muhammad Jakaria Rahimi
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